



OPEN Estimating hip impact velocity and acceleration from video-captured falls using a pose estimation algorithm

Reese Michaels¹, Tiago V. Barreira¹, Stephen N. Robinovitch², Jacob J. Sosnoff³ & Yaejin Moon¹  

Analyzing video footage of falls in older adults has emerged as an alternative to traditional lab studies. However, this approach is limited by the labor-intensive process of manually labeling body parts. To address this limitation, we aimed to validate the use of the AI-based pose estimation algorithm (OpenPose) in assessing the hip impact velocity and acceleration of video-captured falls. We analyzed 110 videos of 13 older adults (64.0 ± 5.9 years old) falling sideways in an experimental setting. By applying OpenPose to each video, we generated a time series of hip positions in the video, which were then analyzed using custom MATLAB code to calculate hip impact velocity and acceleration. These calculations were compared against ground truth measurements obtained from motion capture systems (VICON for hip impact velocity) and inertial measurement units (MC10 for hip impact acceleration). We examined the agreement between the ground truth and OpenPose measurements in terms of mean of absolute error (MAE), mean of absolute percentage error (MAPE), and bias (mean of error). Results showed that OpenPose had a good accuracy in estimating hip impact velocity with minimal bias (MAE: 0.17 ± 0.13 m/s, MAPE: $7.28 \pm 5.21\%$; percent bias: -1.27%). However, its estimation of hip impact acceleration (i.e., peak vertical hip acceleration at impact) showed poor accuracy (MAPE: $26.3 \pm 19.4\%$), showing substantial underestimation in instances of high acceleration impacts (> 3.0 g). Further ANOVA analysis revealed OpenPose's ability to discern significant differences in hip impact velocity and acceleration based on the movement response utilized during the fall (e.g., stick-like fall, tuck-and-roll, knee block). This is the first study to validate the use of a pose estimation algorithm for identifying the hip impact kinematics in video-captured falls among older adults. Future validation studies involving diverse camera settings, fall contexts, and biomechanical parameters are warranted to extend this support for using pose estimation algorithms in this field.

Keywords Pose estimation, Artificial Intelligence, Fall, Biomechanics, Older adults

Falls are the leading cause of accidental injury and death among the elderly¹, causing 90% of hip fractures and 80% of hospital admissions from traumatic brain injury in older adults^{2,3}. The direct medical cost for fall-related injuries reaches \$19 billion annually in the United States alone⁴. The mortality rate from falls among persons aged ≥ 65 years increased by 31% from 2007 to 2016⁵. In addition, as the population ages, the number of annual fall-related injuries in the United States is expected to increase to 5.7 million by the year 2030⁶.

The direct cause of the injury is a mechanical load, applied to the body upon impact, that exceeds the injury threshold⁷. Consequently, past research employed kinematic and kinetic parameters (e.g., fall velocity, impact force) to identify injury risk during lab-based falling experiments⁸. These studies provided direct quantitative evidence that specific strategies—such as installing compliant flooring⁹, wearing hip protectors¹⁰, or performing protective movements during a fall^{8,11,12}—can significantly reduce the risk of fall-related injuries. However, due to safety concerns, these lab-based human subject experiments often rely on restraint harnesses or involve self-initiated falls by young, healthy individuals⁸. As a result, they may not accurately reflect the characteristics of real-life falls experienced by older adults.

¹Department of Exercise Science, Syracuse University, 150 Crouse Dr, Syracuse, NY 13244, USA. ²Department of Biomedical Physiology and Kinesiology, Simon Fraser University, Burnaby, BC, Canada. ³Department of Physical Therapy, Rehabilitation Science, and Athletic Training, University of Kansas Medical Center, Kansas City, KS, USA. ✉email: yamoon@syr.edu

To overcome this barrier, a few studies analyzed real-life falls in long-term care settings captured by security video cameras^{11,13}. They highlighted the situational context of real-life falls, such as activities during the fall, cause of imbalance, and movement response during a fall, as analyzed through visual evaluation of participants' behaviors^{11,13}. Because of its observational nature, however, they were limited in quantifying the biomechanical features of falls. Only one study documented the hip impact velocities of the video-captured real-life falls in older adults¹⁴. This quantitative analysis of movement dynamics during real-life falls offers insights beyond those provided by visual evaluation. However, the time- and labor-intensive process of manually digitizing anatomical landmarks in each video frame hinders its application in large-scale studies or clinical practice.

The rise of pose estimation algorithms presents a promising approach to analyze kinematics in video-captured falls efficiently. The human pose detection algorithm is a recently developed computer vision and artificial intelligence (AI) technique that automatically detects and tracks the position and orientation of human body parts in videos recorded with everyday household devices^{15,16}. Such ease of use holds important promise for capturing movement kinematics, while keeping costs, time, and effort to a minimum^{15,16}. This technology has proven effective in various healthcare contexts, including studying gait characteristics^{17,18} and assisting in other fitness-related tasks¹⁹, demonstrating high accuracy in these measures. Despite its potential, the application of this technology to fall analysis remains largely unexplored. While few studies have utilized this technology for fall detection^{20,21}, their findings only offer a binary result: either a fall happened, or it didn't. No research to date has explored the intricate time-series kinematic data that take place during a fall using the AI-based pose-estimation algorithms.

While the use of pose estimation algorithms shows promise for extracting biomechanical features from video-captured falls, several challenges arise when applying these systems to this context. The extensive range of joint articulations combined with the rapid movements of a fall can make capturing clear videos of such quick actions difficult, especially considering the relatively slow sampling rates and low resolution of many video recording devices^{16,22}. Moreover, since pose estimation algorithms rely on pre-trained models designed mostly for gait analysis, applying them to videos that involve different types of movements—like those seen when people fall—has proven challenging^{16,23}.

In this study, we validated using a pose estimation algorithm, OpenPose, to analyze planar videos capturing a person's falls. OpenPose is a widely used, open-source framework and high-performing, neural network-based software for pose estimation¹⁶. We focused on estimating hip impact severity parameters (i.e., hip velocity and acceleration) from sideways falls since hip fractures are the most common serious injury from elderly falls²⁴. Our objectives were threefold: (1) to assess the validation metrics (mean absolute error, mean absolute percentage error, and bias) of OpenPose in estimating hip impact velocity and acceleration from planar videos capturing falls; (2) to assess whether the validation metrics were influenced by characteristics of the fall, including movement responses (use of tuck-and-roll or knee block movements) or fall directions; (3) to explore whether OpenPose could accurately identify variations in hip impact parameters across different movement responses or fall directions.

Methods

Participants

This is a secondary analysis of the randomized controlled trial of a safe landing training study (NCT#0017577)²⁵. All procedures were approved by the University of Illinois at Urbana-Champaign Institutional Review Board. All experiments were performed in accordance with relevant guidelines and regulations. All participants provided written informed consent prior to participating in the study. The inclusion and exclusion criteria were set to include healthy individuals capable of safely participating in the procedures. To be included, participants had to be: (1) aged between 55 and 75 years, (2) weight between 45 and 100 kg, (3) height between 150 and 195 cm, (4) have a bone mass density t-score above -1.0 , (5) finish a sit-to-stand test 5 times within 10 s, (6) score above 26 on the Montreal Cognitive Assessment (MoCA). Exclusion criteria were having: (1) bone fracture in the past 5 years, (2) a diagnosis of osteoporosis or osteopenia, (3) a history of stroke or neuromuscular conditions, (4) trouble rising from a prone position, (5) pregnancy, and (6) a tendency to bruise easily or delicate skin. These inclusion and exclusion criteria resulted in a sample of physically fit older adults.

Falling experimental setup

During the falling experiments, participants were made to fall sideways onto a 183 cm × 122 cm × 20 cm thick crash pad (Asana Drag Pad, Asana Climbing & MFG, Boise ID, USA) by releasing an inextensible tether that held to the participant at a 10° lean from the vertical position²⁵. The tether was released via a mechanical catch (a snap shackle) and the fall was unpredictably timed by randomly assigning a time delay between 3 and 8 s²⁵. Participants were instructed to “land on the mat in a way that feels comfortable for you”. As the lower-limb impact was the main interest of the primary study, it was emphasized to restrain from breaking the falls with the arms or taking a step²⁵. The test consisted of randomly ordered right-side and left-side falls²⁵.

The participants employed various movement strategies during the fall (i.e., fall strategy), later classified as ‘stick-like fall’, ‘knee block’, or ‘tuck-and-roll’ strategy (Fig. 1). Specifically, a ‘stick-like fall’ referred to instances in the videos where participants neither changed their body orientation from the initial direction of the sideways fall nor blocked the fall with their knees, first impacting the lateral side of their leg or hip. When participants blocked the fall by contacting the ground with their knee well before contacting the ground with their thigh or hip, it was categorized as a ‘knee block’. In the original study, some trials were instructed to perform a ‘tuck-and-roll’ strategy, where participants managed to rotate their bodies backward, landing on the posterior aspect of the hip, and rolling along their backs²⁵.

Video recordings were independently reviewed by two researchers (RM and YM) to determine the fall strategy being executed in each trial. The researchers used the checklist table for classifying the fall strategy

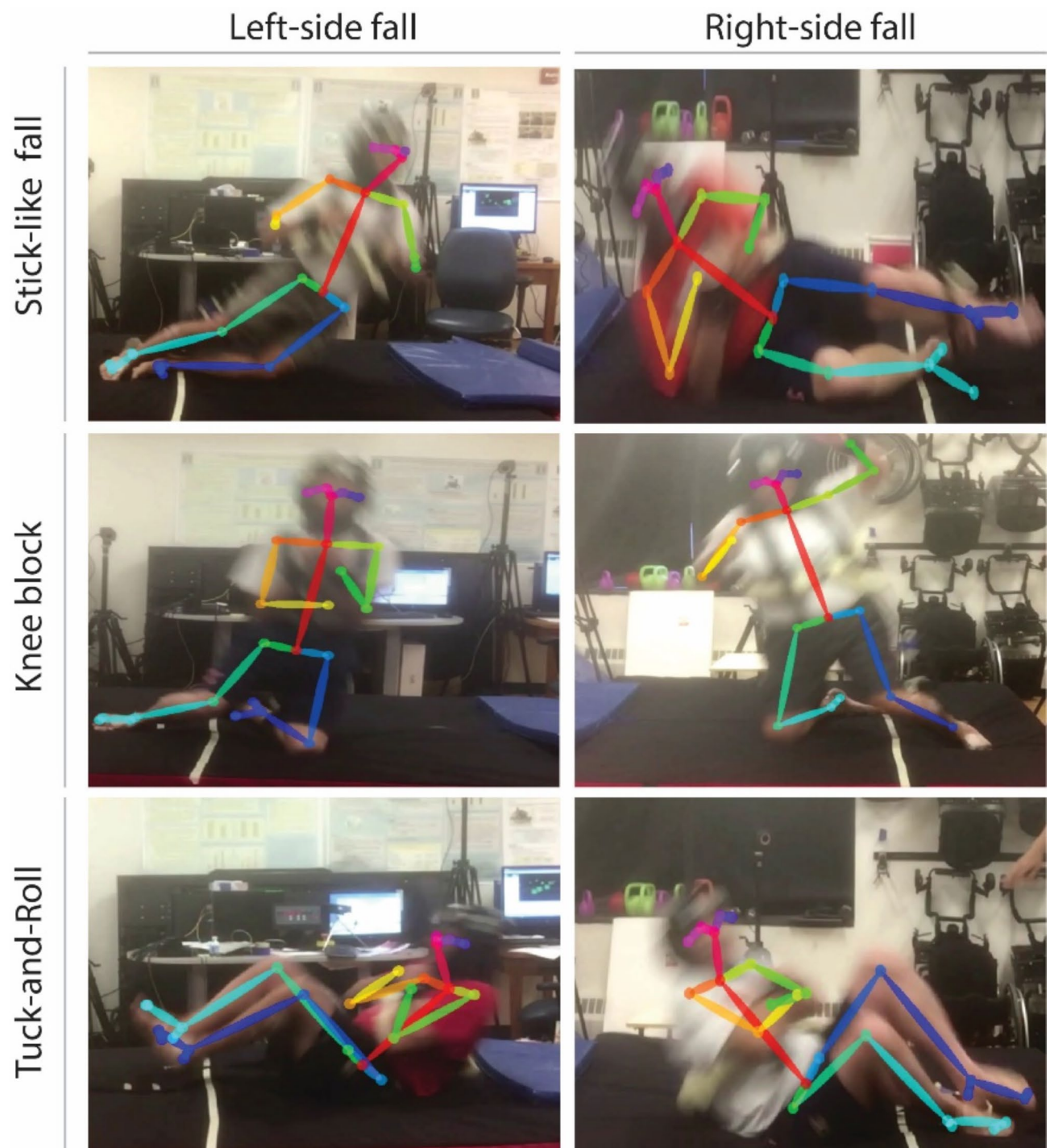


Fig. 1. Example video frames overlaid with key point detection from OpenPose show participants falling either to the left or right side. Participants utilized diverse movement strategies during the fall, such as falling rigidly (described as a ‘stick-like fall’), blocking the fall with their knee (referred to as ‘knee block’), or employing a tuck-and-roll strategy.

(Supplementary Table S1). Out of these 110 trials, only 5 instances (4.5%) showed disagreement between the raters regarding the fall strategy classification, which were resolved through discussion to reach a consensus.

Data collection and analysis for the ground truth measurements

During the falling experiment, hip impact velocity and hip impact acceleration was quantified, as these are the kinematic parameters most frequently reported for estimating the risk of hip fractures⁸. To validate the ability of OpenPose to estimate these measures, a motion capture camera system served as the ground truth for measuring

hip impact velocity, while an inertial measurement unit (IMU) served as the ground truth measure for hip impact acceleration.

The motion capture cameras (Vicon, Oxford Metrics, Oxford, England) were used to collect kinematic data of the hip. Four reflective markers were attached bilaterally to the hip (greater trochanter and posterior iliac spine). The system recorded the 3D positions of these markers at a sampling frequency of 100 Hz. The 3D coordinates of the markers were entered into a VICON Nexus software (VICON, Oxford Metrics, Oxford, England), and hip velocity was calculated. Then, using a custom MATLAB script, hip impact velocity was calculated as the maximum value of hip vertical velocity prior to impact.

Additionally, participants wore a wireless IMU sensor (BioStampRC, MC10, Inc., Lexington, MA, USA)²⁶, on the lower back (L4) to estimate hip impact acceleration. The wireless inertial sensor was utilized since hip impact acceleration was difficult to resolve using motion capture camera system due to marker occlusion at impact during a fall. The device sampled data at 125 Hz. The acceleration measured in the inertial sensor was resolved into components associated with a global frame with a custom-made MATLAB code²⁷. Then, the converted 3D acceleration was digitally filtered with forth-order, zero-phase, low-pass Butterworth filter with a 15 Hz cutoff frequency. Finally, the hip impact acceleration was identified as maximum hip vertical acceleration with a custom MATLAB script. The validity of the IMU sensor (BioStampRC, MC10) was examined in a previous gait analysis study²⁶, and its accuracy in measuring the hip impact acceleration was internally validated (Supplementary Fig. S1).

Video recording

A digital RGB video camera was placed to record the frontal plane view of the participant when they were standing still prior to the release of the tether for the fall experiment. The plane of view changed if the participant rotated their body during the fall. The video was recorded at 30 frames per second (fps) and had a resolution of 1920 × 1080 pixels. The height of the camera placement was set to about 110 cm from the ground, aligning with the participant who stood approximately 45 cm off the floor due to the combined thickness of the platform and crash pad. The captured videos were cropped to exclude anyone other than the participant from the view.

OpenPose video analysis process

Video recordings were processed with OpenPose using the BODY_25 keypoint model that tracks the following 25 keypoints: nose, neck, midhip and bilateral keypoints at eyes, ears, shoulders, elbows, wrists, hips, knees, ankles, heels, big toes and small toes. Additionally, we used OpenPose flag to limit the maximum number of people to one person, as we had cropped the video to include only the participant in the frame. The OpenPose analysis outputted JSON files for each frame of the video, containing the pixel coordinates of each detected key point. These JSON files were then downloaded for more detailed analysis in a custom MATLAB script. We modified the MATLAB code available from GitHub (<https://github.com/janstenum/GaitAnalysis-PoseEstimation>), which was developed to analyze spatio-temporal gait parameters¹⁷, to extract hip impact velocity and hip impact acceleration from the videos. The steps of the MATLAB processing are (Fig. 2):

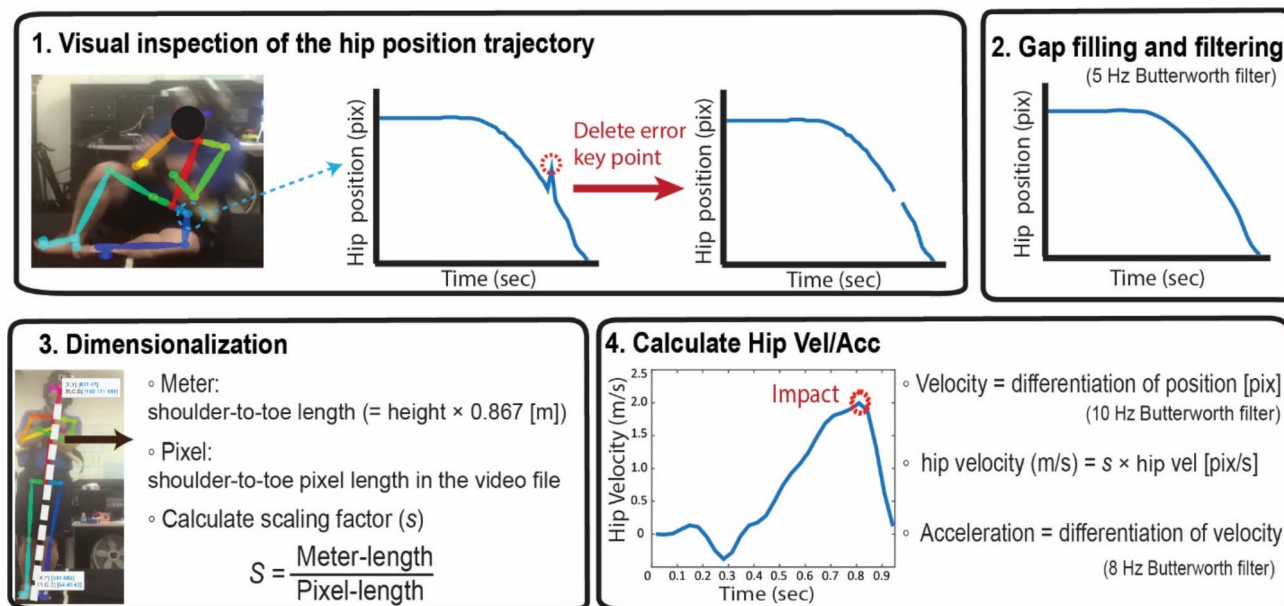


Fig. 2. Flow chart of MATLAB processing. The hip position data recorded in the video was obtained from OpenPose analysis and was processed through four steps in MATLAB to calculate hip impact velocity and acceleration.

Step 1. Visual inspection of trajectory

When OpenPose erroneously switched the left–right identification of the limbs, we visually inspected the trajectories of the left and right hip key point to identify and correct the key points in these frames. We also deleted key points from the frames that erroneously identified the body key points.

Out of the 110 video files, 14 videos (12.7%) required hip key point modification due to erroneous positioning. Of these, 9 videos required only a single-frame key point correction. A total of 4530 frames were analyzed across all video recordings, and only 27 frames (0.6% of the total frames) required correction of the hip key point.

Step 2. Gap filling and filtering

We filled gaps in key point position trajectories (i.e., frames where OpenPose did not detect all key points) using linear interpolation for gaps spanning up to four video frames. We then filtered the position data using a zero-lag 4th order low-pass Butterworth filter with a cutoff frequency at 5 Hz¹⁷.

Step 3. Obtaining dimensionalized coordinate values from the pixel coordinates

To generate estimates of hip velocity and acceleration that incorporate spatial information, it was necessary to scale the video to obtain dimensionalized coordinate values from the pixel coordinates. Previous studies using pose-estimation algorithms accomplished this by scaling (s) the video to known measurements on the ground using the following equation as:

$$s = \frac{\text{Distance}}{\text{Pixel length}}$$

This could be achieved by placing an object of known size in the field of view at the same depth or using a known anthropometric quantity such as body height¹⁷. In this study, although we obtained height information from the participants, some of the camera views cut off the head in the full standing position, limiting our ability to utilize the height data. Therefore, we incorporated the anthropometric ratio of shoulder-to-toe length with respect to height, which is the distance multiplied by 0.867 times the height²⁸. The pixel length was calculated by measuring the length between the shoulder coordinate and the toe coordinate, in pixels, within the video's field of view.

Step 4. Calculating hip vertical velocity and acceleration

Since we consider vertical hip velocity, only the vertical component (i.e., y_{pixel}) of the right and left hip keypoints was considered for further calculation. We first calculated vertical hip velocity in pixel dimension (i.e., $v_{y\text{-pixel}}$) by computing numerical differentiation of the y_{pixel} . This was then filtered using a zero-lag, 4th order low-pass Butterworth filter with a cutoff frequency at 10 Hz²². Then, we calculated dimensionalized vertical hip velocity as:

$$\text{Vertical hip velocity} = s \times v_{y\text{-pixel}}$$

Next, to calculate the vertical hip acceleration, we performed a numerical differentiation of the vertical hip velocity. We then refined the resulting data by applying a zero-lag, fourth-order low-pass Butterworth filter with a cutoff frequency of 8 Hz. This frequency was selected based on empirical sensitive analysis comparing the accuracy of OpenPose data in estimating hip impact acceleration across eight different cutoff frequencies: 5 Hz, 6 Hz, 7 Hz, 8 Hz, 9 Hz, 10 Hz, 11 Hz, and 12 Hz (see Supplementary section 'Sensitivity Analysis of Cutoff Frequency for Acceleration Calculations in OpenPose Data'). When compared to ground truth data from an IMU, the error was lowest at 8 Hz (26% error). Based on this analysis, we determined that 8 Hz is the most appropriate lower bound cutoff frequency for capturing hip impact acceleration during falls using the OpenPose estimation²².

Finally, the hip impact velocity was identified as the maximum value of the vertical hip velocity (Fig. 3). The hip impact acceleration was identified as the maximum value of the vertical hip acceleration within five frames of the visually identified frame where the initial hip impact occurred.

Statistical analysis

All statistical analyses were performed with IBM SPSS statistics for Windows (version 27; IBM SPSS Inc., Armonk, NY, USA). For all ANOVA tests, normality of residuals was tested using Shapiro–Wilk test. In case of significant main effects in ANOVA analysis, LSD post-hoc analyses were carried out for pair-wise comparisons. Also, measure of eta-squared (η^2) was obtained as the effect size for ANOVA analyses. All analyses used two-sided tests, and the alpha level was set at 0.05.

Differences in hip impact parameters among fall contexts

First, we explored characteristics of hip impact velocity and acceleration, as recorded by the ground truth measures, across various fall contexts. These included the movement strategies utilized during a fall (i.e., fall strategies such as stick-like falls, knee blocks, tuck-and-roll) and fall directions (left vs. right). To achieve this, we performed two-way ANOVA (Fall Strategy $\times$ Fall Side) analyses on each hip impact parameter, as estimated by the ground truth measures.

Validation metrics

To validate OpenPose estimations against the ground truth measures, we calculated mean of the absolute error (MAE), which is the absolute magnitude of the difference between the OpenPose estimations and those measured by the ground truth ($= |\text{OpenPose} - \text{Ground truth}|$)²⁹. Additionally, mean of the absolute percentage error (MAPE) was calculated, representing the percentage of the absolute error between the OpenPose estimation and the ground truth value with respect to the ground truth value ($= \frac{|\text{OpenPose} - \text{Ground truth}|}{\text{Ground truth}} \times 100\%$)²⁹.

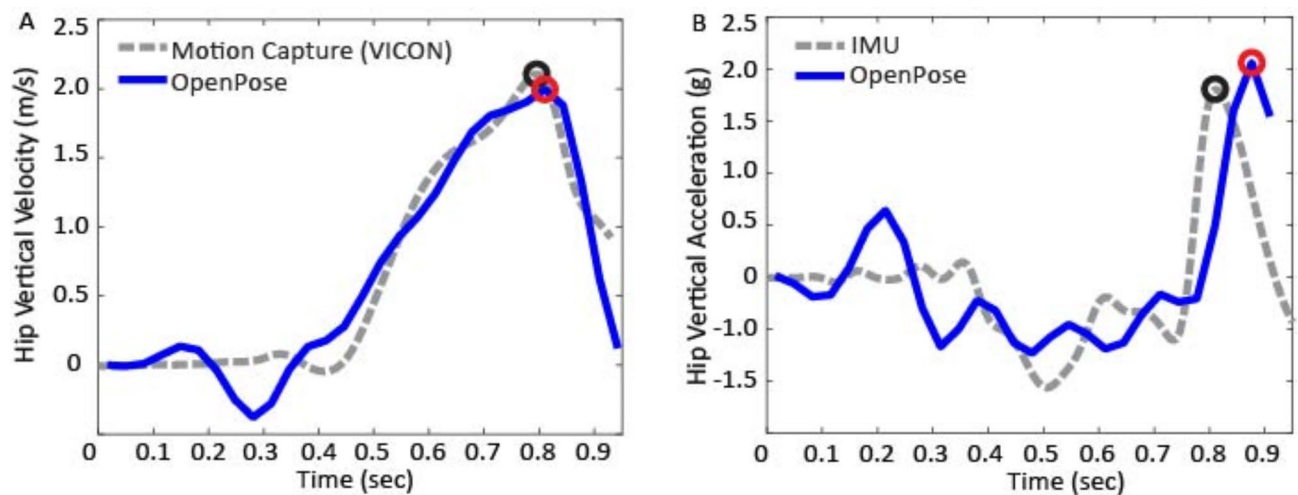


Fig. 3. Time profile of hip velocity and acceleration for a representative participant in a single sideways fall trial, captured concurrently using Motion Capture (VICON), IMU (MC10), and a video analyzed by OpenPose. **(A)** Hip vertical velocity estimated by motion capture (ground truth) and measured by OpenPose. A red and a black circle indicate the hip impact velocity estimated by motion capture (black) and by OpenPose (red), respectively. **(B)** Hip vertical acceleration estimated by IMU (ground truth) and measured by OpenPose. A red and a black circle indicate the hip impact acceleration estimated by IMU (black) and by OpenPose (red), respectively.

Additionally, we identified biases, which pertain to the scale and orientation (i.e., over or underestimations) of the errors²⁹. Bias was defined as the mean of the errors between the OpenPose estimation and that measured by the ground truth measures ($= \text{OpenPose} - \text{Ground truth}$), whereas percentage bias was the mean of the percentage errors ($= \frac{\text{OpenPose} - \text{Ground truth}}{\text{Ground truth}} \times 100\%$).

For the MAPE and percentage bias, a classification system was used to assess acceptable standards of errors, where a percent error $\leq 5\%$ indicates excellent agreement, $5\% < \text{percent error} \leq 10\%$ indicates good agreement, $10\% < \text{percent error} \leq 15\%$ indicates acceptable agreement, and percent error $> 15\%$ indicates poor agreement^{30,31}.

Bland–Altman plots were generated for visually examining systematic patterns of error in estimation of the hip impact parameters. Pearson's correlation analysis was conducted to examine the relationship between the OpenPose and ground truth measures. Conventionally, Pearson's correlation coefficient (r) of 0.4, 0.6, and 0.8 are considered to represent fair, moderate, and strong correlations, respectively³².

Comparison of validation metrics among fall contexts

We also investigated whether the validation metrics, MAE and Bias, exhibited differences across various fall contexts (i.e., movement strategies and fall directions), by conducting two-way ANOVA (Fall Strategy $\times$ Fall Side) analyses on each validation metric.

Discriminative validity of OpenPose

Furthermore, we validated ability of OpenPose to accurately detect differences in hip impact parameters across various fall contexts (i.e., movement strategies and fall directions), comparing these findings to the ground truth results. To achieve this, we performed two-way ANOVA (Fall strategy $\times$ Fall Side) analyses on each hip impact parameter as estimated by OpenPose.

Results

Participant characteristics

Table 1 presents the participants' demographic information. A total of 13 participants underwent the study procedures. The participants' ages ranged from 55 to 73 years, with an average of 64.0 ± 5.9 years. Ten of the 13 participants were male (77%). All participants were physically and cognitively healthy, as indicated by their bone mass density, 5 times sit-to-stand test, and MoCA scores.

Characteristics of movement responses during a fall (i.e., fall strategy)

A total of 110 videos recording sideways falls were analyzed. Among these, 42 videos (38%) recorded falls toward the left side, while 68 videos (62%) recorded falls to the right side. Since the use of upper limb braces was avoided as per instructions, all participants landed on the ground, with either their legs or hips making the initial impact. Specifically, in 27 videos (25%), participants fell using a stick-like fall strategy. In 40 videos (36%), participants demonstrated a knee block strategy, and in 46 videos (39%), participants employed a tuck-and-roll strategy (see “Falling experimental setup” section for descriptions of each fall strategy).

Characteristic	Mean ± SD
N	13 (3F/10M)
Age (years)	64.0 ± 5.9
Height (cm)	171.1 ± 7.9
Weight (kg)	73.6 ± 13.0
Bone mass density (t-score)	0.17 ± 0.96
5 times sit to stand (sec)	7.60 ± 1.2
MoCA	28.3 ± 1.3

Table 1. Participants demographics.

Parameters	Measurement	Side/fall strategy	Mean	S.D	S.E	Min	Max
Hip impact velocity (m/s)	Motion capture	All (N = 110)	2.32	0.36	0.03	1.61	3.22
		Fall side					
		Left fall (N = 42)	2.30	0.41	0.06	1.61	3.22
		Right fall (N = 68)	2.33	0.33	0.04	1.81	3.05
		Fall strategy					
		Stick-like fall (N = 27)	2.76	0.30	0.06	2.04	3.22
		Knee block (N = 40)	2.05	0.17	0.03	1.61	2.38
		Tuck-and-roll (N = 43)	2.29	0.25	0.04	1.73	2.73
Hip impact acceleration (g)	IMU accelerometer	All	2.15	0.95	0.09	0.87	5.15
		Fall side					
		Left fall	2.27	1.14	0.18	0.87	5.15
		Right fall	2.07	0.82	0.10	0.87	4.28
		Fall strategy					
		Stick-like fall	3.25	0.84	0.16	1.96	5.15
		Knee block	1.46	0.43	0.07	0.92	2.66
		Tuck-and-roll	2.09	0.72	0.11	1.73	2.73

Table 2. Characteristics of hip impact parameters recorded by the ground truth measurements.

Characteristics of hip impact velocity and acceleration

Kinematic hip impact parameters, measured by gold-standard equipment, exhibited a broad range across participants and the movement strategies they employed during a fall and the direction of the fall (Table 2). All hip impact parameters were significantly different among the fall strategies (hip impact velocity: $F(2,104) = 72.0$, $p < 0.01$, $\eta^2 = 0.57$; hip impact acceleration: $F(2,104) = 57.8$, $p < 0.01$, $\eta^2 = 0.52$). Post-hoc analysis revealed that, for both hip impact parameters, all strategies exhibited significantly different values from one another: the stick-like fall exhibited the highest value ($p < 0.01$, 95% CI [0.33 0.62]), followed by the tuck-and-roll strategy ($p < 0.01$, 95% CI [0.11 0.37]), with the knee block strategy exhibiting the lowest value ($p < 0.01$, 95% CI [− 0.86 − 0.57]). However, there was no difference in these hip impact parameters between the right and left side of the falls (all p values > 0.05).

MAE, MAPE, and bias of hip impact velocity estimated by OpenPose

Overall, the MAE of hip impact velocity estimated by OpenPose was 0.17 ± 0.13 m/s, when compared to the ground truth measures (Table 3). The MAPE of the estimate was $7.28 \pm 5.21\%$. The correlation analysis also showed that there was strong correlation between the hip impact velocity estimated by the OpenPose and the ground truth measure of VICON ($r = 0.83$, $p < 0.01$; Fig. 4A).

There was minimum bias in the error with an average of -0.04 m/s as the bias, and -1.27% as the percent bias. The Bland–Altman plot confirms this observation showing that the estimated hip impact velocity was evenly spread out between over or underestimations compared to the reference (Fig. 4B).

Next, we examined whether these validity indices differed among various fall contexts (i.e., fall sides and fall strategies). For MAE, there was no significant mean difference between the fall sides ($p = 0.51$, $\eta^2 < 0.01$), among the fall strategies ($p = 0.50$, $\eta^2 < 0.01$), nor was there an interaction effect between fall side and fall strategy ($p = 0.69$, $\eta^2 < 0.01$). In bias, there was a significant mean difference in moderate effect size among the fall strategies ($F(2,104) = 5.35$, $p = 0.01$, $\eta^2 = 0.09$). However, it should be noted that the bias for all strategies was less than 5% for either over- or underestimated values.

MAE, MAPE, and bias of hip impact acceleration estimated by OpenPose

The MAE of hip impact acceleration estimated by OpenPose analysis was 0.57 ± 0.53 g, when compared to the ground truth measures (Table 4). The MAPE of the estimate was $26.3 \pm 19.4\%$. The correlation analysis showed a

	Mean absolute error (MAE)		Mean absolute percentage error (MAPE)		Bias	Percent bias
	AVG $\pm$ S.D. (m/s)	95% CI (m/s)	AVG $\pm$ S.D. (%)	95% CI (%)	AVG $\pm$ S.D. (m/s)	AVG (%)
All data	0.17 $\pm$ 0.13	[0.14 0.19]	7.28 $\pm$ 5.21	[6.29 8.26]	-0.04 $\pm$ 0.21	-1.27
Fall side						
Left fall	0.16 $\pm$ 0.14	[0.11 0.20]	6.94 $\pm$ 5.66	[5.18 8.71]	-0.05 $\pm$ 0.20	-1.73
Right fall	0.17 $\pm$ 0.12	[0.14 0.20]	7.48 $\pm$ 4.94	[6.29 8.68]	-0.03 $\pm$ 0.21	-0.99
Fall strategy						
Stick-like fall	0.17 $\pm$ 0.16	[0.11 0.23]	6.07 $\pm$ 5.30	[3.97 8.17]	-0.11 $\pm$ 0.21	-3.70
Knee block	0.15 $\pm$ 0.10	[0.11 0.18]	7.20 $\pm$ 4.96	[5.58 8.75]	+0.04 $\pm$ 0.17	+2.03
Tuck-and-roll	0.19 $\pm$ 0.16	[0.15 0.23]	8.14 $\pm$ 5.34	[6.50 9.78]	-0.07 $\pm$ 0.22	-2.81

Table 3. Validity indices (MAE, MAPE, and bias) of hip impact velocity. S.D. Standard deviation, 95% CI 95% confidence interval.

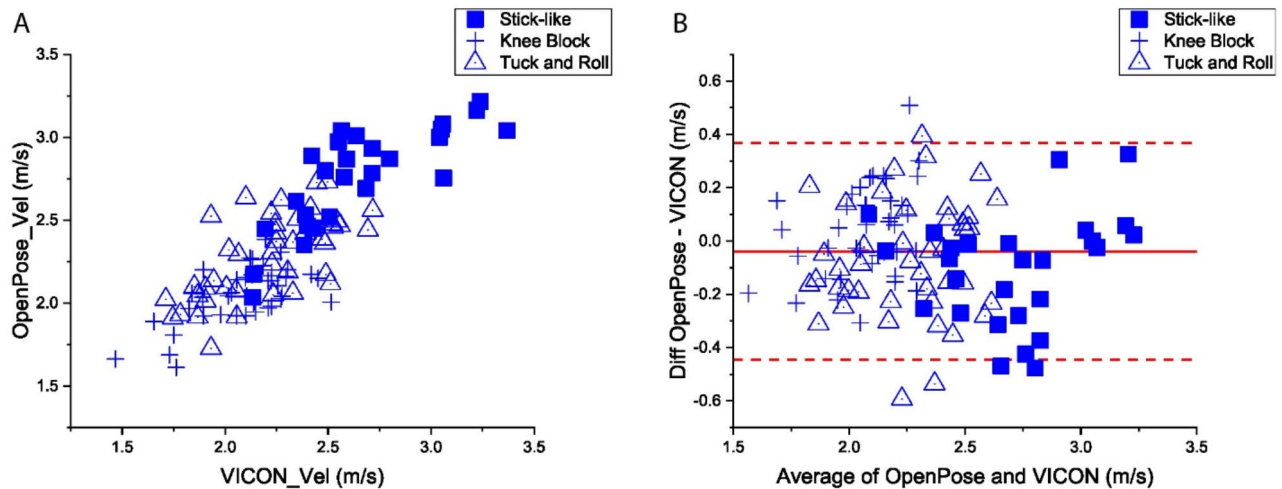


Fig. 4. Hip impact velocity estimated by OpenPose compared to the ground truth (VICON motion capture). (A) A scatter plot showing the strong correlation ($r=0.83$) between the hip impact velocity estimated by OpenPose (y-axis) and that yielded by VICON motion capture (x-axis, ground truth). (B) A Bland–Altman plot of hip impact velocity by OpenPose compared with that of the ground truth measure (VICON motion capture). Limits of agreement are specified as the average difference (red solid line) and 95% confidence interval of the error (red dotted line). Each dot on both graphs represents a single video in the test set, grouped by the fall strategy utilized during the fall.

	Mean absolute error (MAE)		Mean absolute percentage error (MAPE)		Bias	Percent bias
	AVG $\pm$ S.D. (g)	95% CI (g)	AVG $\pm$ S.D. (%)	95% CI (%)	AVG $\pm$ S.D. (g)	AVG (%)
All data	0.57 $\pm$ 0.53	[0.47 0.67]	26.3 $\pm$ 19.4	[22.7 30.0]	-0.31 $\pm$ 0.71	-5.21
Fall side						
Left fall	0.65 $\pm$ 0.70	[0.43 0.86]	25.0 $\pm$ 17.3	[19.6 30.4]	-0.46 $\pm$ 0.84	-10.2
Right fall	0.52 $\pm$ 0.38	[0.43 0.61]	27.2 $\pm$ 20.7	[22.2 32.2]	-0.22 $\pm$ 0.61	-2.14
Fall strategy						
<i>Stick-like fall*</i>	<i>0.91 $\pm$ 0.79*</i>	<i>[0.60 1.22]</i>	25.3 $\pm$ 16.4	[18.9 31.9]	<i>-0.81 $\pm$ 0.90*</i>	<i>-21.4</i>
Knee block	0.40 $\pm$ 0.30	[0.30 0.50]	27.4 $\pm$ 21.5	[20.5 34.3]	-0.03 $\pm$ 0.50	+4.06
Tuck-and-roll	0.51 $\pm$ 0.38	[0.39 0.62]	26.0 $\pm$ 19.5	[20.0 32.0]	-0.25 $\pm$ 0.58	-3.68

Table 4. Validity indices (MAE, MAPE, and bias) of hip impact acceleration. Italics and asterisks(*) indicate significantly poorer performance of OpenPose in Stick-like Fall compared to other fall strategies. S.D. Standard deviation, 95% CI 95% confidence interval.

moderate correlation between the hip impact acceleration estimated by OpenPose and the ground truth measure of IMU ($r=0.67$, $p<0.01$; Fig. 5A).

The bias results showed that OpenPose tended to underestimate the hip impact acceleration compared to the ground truth, with an average bias of -0.31 g and a percent bias of -5.21% . The Bland–Altman plot further revealed that the underestimation occurred when the hip impact acceleration exceeded 3.0 g, with 6 data points falling below the lower-bound of the 95% confidence interval (CI) for the error in this region (Fig. 5B). All these data points were associated with a stick-like fall.

For the MAE of hip impact acceleration, there was a significant difference among the fall strategies ($F(2,104)=11.1$, $p<0.01$, $\eta^2=0.18$) and in the interaction effect of fall side $\times$ fall strategy ($F(2,104)=5.68$, $p=0.01$, $\eta^2=0.10$), but not between fall side ($p=0.16$, $\eta^2=0.02$). Post-hoc analysis revealed that, particularly in the case of a left-side fall, the stick-like fall strategy (MAE: 1.22 ± 0.97 g) had significantly higher error compared to the knee block (MAE: 0.37 ± 0.34 g; $p<0.01$, 95% CI [0.40 1.30]) and the tuck-and-roll strategy (MAE: 0.40 ± 0.27 g; $p<0.01$, 95% CI [0.36 1.30]). The right-side fall did not show this difference in accuracy among the fall strategies (all p values >0.05).

For bias, there was a significant difference among the fall strategies ($F(2,104)=13.2$, $p<0.01$, $\eta^2=0.20$), and in the interaction effect of fall side $\times$ fall strategy ($F(2,104)=4.94$, $p=0.01$, $\eta^2=0.09$). There was also significant difference between the fall sides, but the effect size was small ($F(1,104)=3.85$, $p=0.05$, $\eta^2=0.04$). Post-hoc analysis revealed that, only in the left-side fall, the stick-like fall strategy had a significantly greater underestimated bias (average: -1.23 ± 0.97 g) compared to the knee block (average: -0.10 ± 0.50 g; $p<0.01$, 95% CI [$-1.64 - 0.62$]) and the tuck-and-roll strategy (average: -0.12 ± 0.48 g; $p<0.01$, 95% CI [$-1.64 - 0.57$]). As shown in the Bland–Altman plot (Fig. 5B), this trend is linked to the fact that the stick-like fall had the highest hip impact acceleration (x-axis), and the error (y-axis) increased with increased impact acceleration.

Discriminative validity of OpenPose

Next, we explored whether OpenPose could accurately discriminate variations in hip impact parameters across different fall strategies and sides. We evaluated how well these estimates from OpenPose correspond with findings from ground truth measurements (see “Characteristics of hip impact velocity and acceleration” section, Characteristics of Hip Impact Velocity and Acceleration).

OpenPose estimation revealed significant differences in hip impact parameters across the different fall strategies (Fig. 6A,C; hip impact velocity: $F(2,104)=36.7$, $p<0.01$, $\eta^2=0.41$; hip impact acceleration: $F(2,104)=36.5$, $p<0.01$, $\eta^2=0.41$). Post-hoc analysis revealed significant differences in both hip impact velocity and acceleration across all fall strategies. For hip impact velocity, the stick-like fall strategy had the highest velocity ($p<0.01$, 95% CI [0.43, 0.70]), followed by the tuck-and-roll strategy ($p<0.01$, 95% CI [0.30, 0.57]), while the knee block strategy showed the lowest values ($p=0.04$, 95% CI [$-0.25, -0.01$]). Hip impact acceleration followed the same trend, with the stick-like fall strategy showing the highest acceleration ($p<0.01$, 95% CI [0.77, 1.23]), followed by the tuck-and-roll strategy ($p<0.01$, 95% CI [0.37, 0.82]), while the knee block strategy showed the lowest values ($p<0.01$, 95% CI [$-0.61, -0.20$]). However, there was no significant difference in the hip impact parameters between the falling side (Fig. 6B,D; hip impact velocity: $F(1,104)=3.36$, $p=0.07$, $\eta^2<0.03$;

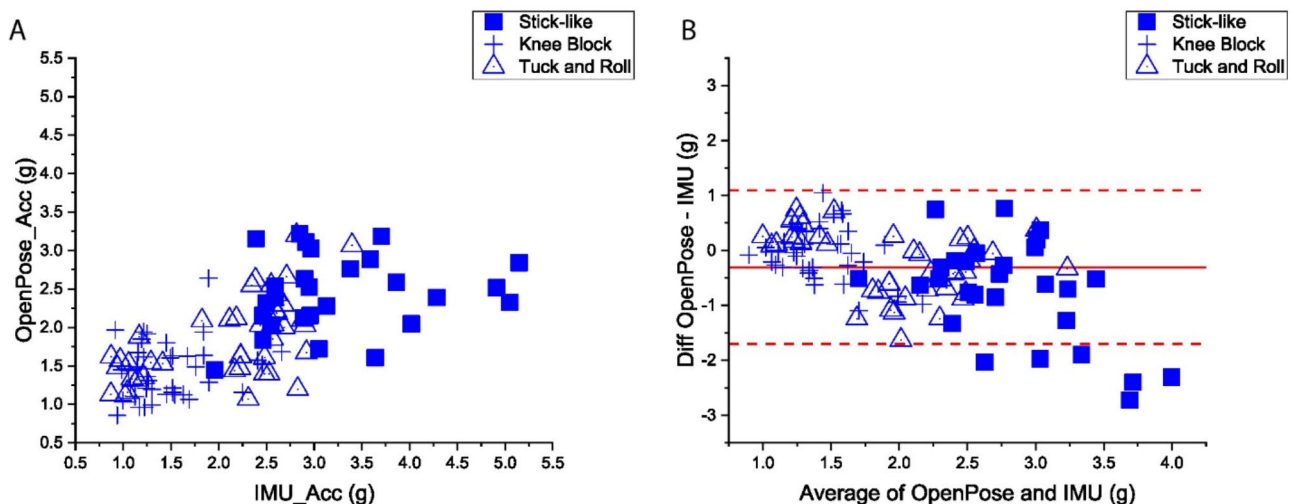


Fig. 5. Hip impact acceleration estimated by OpenPose compared to the ground truth (IMU). **(A)** A scatter plot showing the moderate correlation ($r=0.67$) between the hip impact acceleration estimated by OpenPose (y-axis) and that yielded by IMU (x-axis, ground truth). **(B)** A Bland–Altman plot of hip impact acceleration by OpenPose compared with that of the ground truth measure (IMU). Limits of agreement (LoA) are specified as the average difference (red solid line) and 95% confidence interval of the error (red dotted line). Each dot on both graphs represents a single video in the test set, grouped by the fall strategy utilized during the fall.

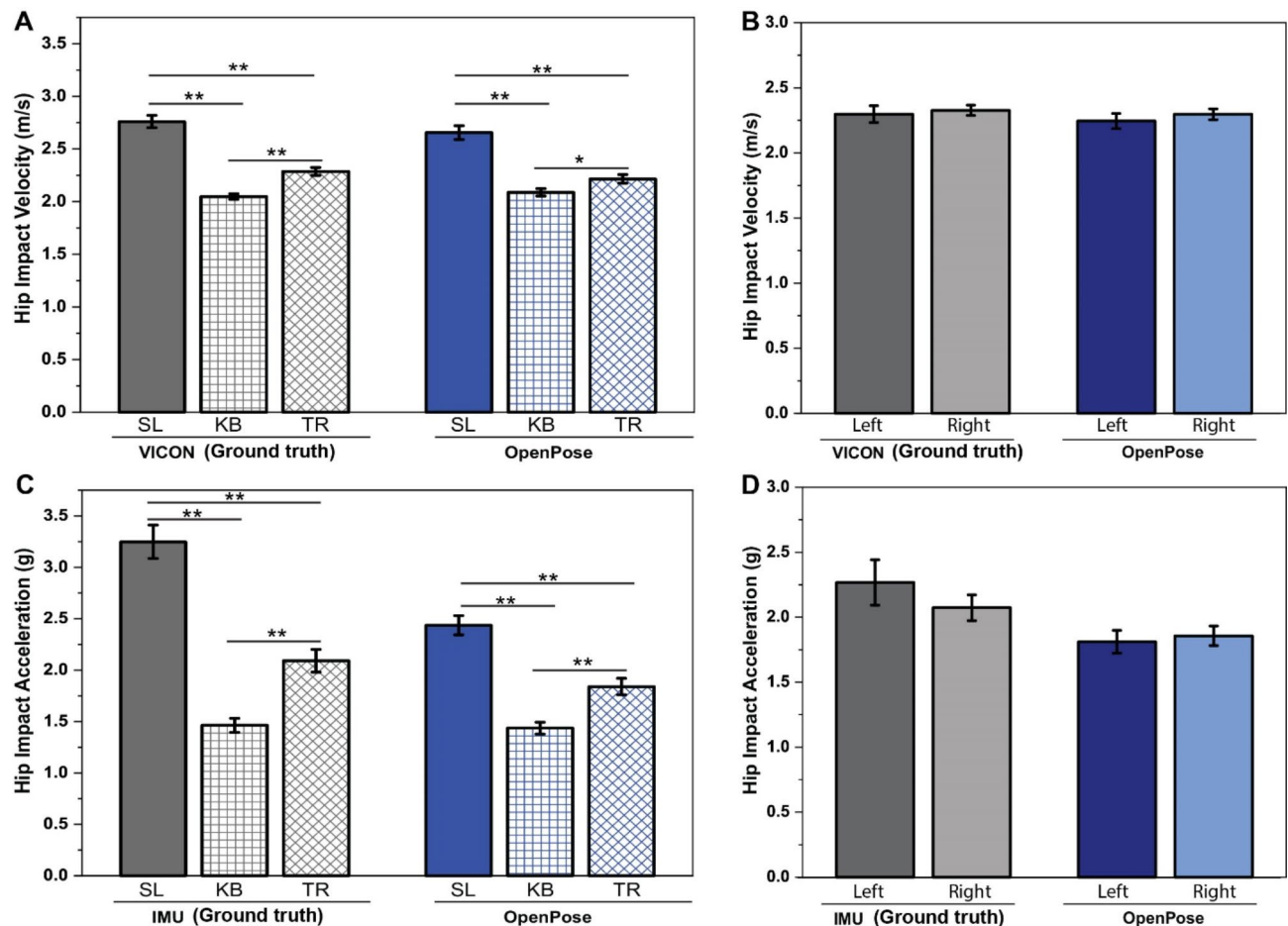


Fig. 6. Comparative analysis of hip impact parameters across fall strategies and sides: Ground truths versus OpenPose. (A,B) Comparison of hip impact velocity across fall strategies (Panel A) and between fall sides (Panel B) measured by VICON (ground truth) and OpenPose. (C,D) Comparison of hip impact acceleration across fall strategies (Panel C) and between fall sides (Panel D) measured by IMU (ground truth) and OpenPose. SL stick-like fall, KB knee block, TR tuck-and-roll; * $p < 0.01$, ** $p < 0.05$.

hip impact acceleration: $F(1,104) = 1.58$, $p = 0.21$, $\eta^2 = 0.02$). All these results of differences were consistent with the ground truth measurement.

Discussion

The primary objective of this study was to assess the validation metrics (MAE, MAPE, bias) of video-based pose estimation using OpenPose in estimating kinematic hip impact parameters (i.e., hip impact velocity and acceleration) during video-captured experimental falls in older adults ($N = 13$, aged 64.0 ± 5.9 years). Our results indicated that the pose estimation algorithm achieved good accuracy in estimating hip impact velocity (i.e., peak vertical hip velocity), with an MAPE of $7.28 \pm 5.21\%$. However, the estimation of hip impact acceleration (i.e., peak vertical hip acceleration at impact) showed poor accuracy, with MAPE of $26.3 \pm 19.4\%$. Notably, there was a substantial underestimation of this measure when participants, employing a stick-like fall response, impacted the ground with high acceleration (> 3.0 g). Furthermore, we observed that the pose estimation algorithm could accurately distinguish characteristics of hip impact parameters based on the movement response utilized during the fall. To our knowledge, this is the first study to validate the use of a pose estimation algorithm for identifying the kinematic parameters in video-captured falls among older adults.

Our results showed that OpenPose estimation of the hip impact velocity demonstrates a good agreement (MAE: 0.17 ± 0.13 m/s, MAPE: $7.28 \pm 5.21\%$) and minimum bias (-0.04 m/s, -1.27%) against ground truth measures^{30,31}. These findings closely align with a previous study that found hip impact velocity could be accurately estimated in a video-captured experimental fall (0.12 m/s, 3.9% average error) using a traditional video analysis software, Kinovea²². It is important to highlight that Kinovea, the software used in the previous study, does not employ advanced computer vision or AI algorithms for detecting body key points. Instead, it depends significantly on the manual labeling of the body key points in each frame of the video. In contrast, OpenPose required minimal human involvement in correcting hip key point detection errors, with adjustments needed on only 0.6% of total frames analyzed (27 frames out of 4530 frames). This limited correction workload significantly reduced time and human labor resources, underscoring the robustness of the algorithm. The minimal nature of

these corrections also suggests that the risk of introducing bias through the correction process was exceedingly low, contributing to the reliability of our analysis.

Previous studies in gait analysis have applied diverse pose estimation algorithms (e.g., OpenPose, DeepLabCut) and reported a percentage error of 3–4% in estimating gait velocity^{17,18}. It is important to note that the falling velocity measured in our study (average: 2.32 m/s) is 2–3 times greater than the average gait velocity reported in previous studies (healthy individuals¹⁷: 1 m/s; stroke population¹⁸: 0.68 m/s). Despite the rapid movement during a fall, which can cause blurred images, OpenPose still achieved acceptable accuracy, with a 7.28% error in estimating key point velocity. Given that the hip impact velocity is the most frequently reported impact severity parameters of a fall⁸, the accuracy and efficiency of the OpenPose in estimating the parameter demonstrate promise for its application in large-scale study to understand injury biomechanics of a fall.

Another essential observation is that OpenPose's accuracy in estimating the hip impact velocity remained consistent across different fall contexts, such as the direction of the fall and the movement strategy employed. This finding is particularly noteworthy considering that the anatomical plane of view is a crucial factor that could impact the accuracy of OpenPose's estimates¹⁶. Specifically, when a participant performed the tuck-and-roll strategy, their body rotation during a fall altered the plane of view recorded in the video from frontal to sagittal view. Potentially, we did not observe the influence of the fall strategy and subsequent changes in the plane of view on the estimates because hip impact velocity was calculated solely based on the vertical component (i.e., the gravitational axis) of the hip position, which remains unchanged in different views.

The estimation of hip impact acceleration by OpenPose had poor agreement (MAE: 0.57 ± 0.53 g, MAPE: $26.3 \pm 19.4\%$) against ground truth measures. Moreover, there was substantial underestimation when participants employed a stick-like fall movement (21.4% underestimation). Given that a stick-like fall results in the highest hip impact acceleration (average: 3.25 ± 0.84 g), the error in estimating stick-like falls appears to stem from OpenPose's limited capability to estimate hip impact acceleration over 3.0 g from the current video data (Fig. 5). It should be noted that the video data in the current study were recorded at a 30 Hz frame rate, while the ground truth measure of acceleration (IMU) was collected at a higher sampling frequency (125 Hz). Therefore, this underestimation at higher accelerations may be due to the lower frame rate of the video. It is possible that using a higher frame rate for video recording could improve video-based acceleration estimation.

The observation on the hip impact acceleration is particularly important given that in real-world falls, the threshold for accelerometer-based fall detection is often set between 3 and 4 g^{33,34}. Therefore, for real-life falls occurring on rigid surfaces, the impact acceleration value would be higher, and OpenPose may be inaccurate in estimating hip impact acceleration in such scenarios. A potential workaround to overcome this limitation is to combine video-based biomechanical analysis with wearable inertial sensors to track acceleration data during real-life falls. Past research has successfully detected real-world falls using wearable inertial sensors^{34,35}. However, a limitation of these sensors for understanding falls in real-life settings is that they are typically placed on only one body part (e.g., the wrist), which restricts a comprehensive understanding of body movement during a fall. By integrating video-captured data with inertial data, we may gain a more holistic understanding of whole-body movement during a fall while also obtaining accurate acceleration measurements.

In fact, while pose estimation has been widely used to estimate diverse kinematic parameters (e.g., position, velocity, joint angle, angular velocity) in various movement domains, the application of this algorithm to estimate the acceleration of key body points has not been extensively explored¹⁶. Consequently, identifying the source of error in estimating hip impact acceleration becomes challenging. The errors could potentially originate from the low frame rate of the camera settings (30 Hz), or the inherent challenge associated with filtering data when taking second derivatives to calculate acceleration from position data³⁶. Therefore, future research that systematically investigates the origins of these errors, especially in contexts involving rapid acceleration such as injury prevention or sports performance, is strongly recommended.

Further, OpenPose's estimation of hip impact parameters can accurately identify significant differences among fall strategies—the stick-like fall has the highest impact, followed by tuck-and-roll, with the knee block having the lowest. This finding corresponds to a previous study that analyzed real-life falls of older individuals captured in video footage. Specifically, the study reported that rotating from sideways to land backward, similar to the 'tuck-and-roll' strategy in the current study, decreased the odds of serious injury in real life (Odds Ratio = 0.37)¹¹. While these studies on real-life video-captured falls linked injury status to various fall contexts (e.g., fall direction, protective responses, landing configuration)^{11,13}, they did not explore digitized time-series biomechanical parameters to address mechanisms of injury. The current observation showed potential to fill this gap by applying OpenPose to video-captured falls to extract the kinematic movement features to underpin injury mechanisms.

However, it should be noted that the effect sizes of the significant differences among the fall strategies estimated by OpenPose were lower than those of the ground truth measurements for hip impact velocity (OpenPose: $\eta^2 = 0.41$, Ground truth: $\eta^2 = 0.57$) and hip impact acceleration (OpenPose: $\eta^2 = 0.41$, Ground truth: $\eta^2 = 0.52$). Therefore, it is possible that the power to detect significant differences in hip impact severity among different fall contexts could be compromised with OpenPose analysis. Future studies are warranted to improve the accuracy of measuring impact parameters extracted with OpenPose, to enhance the clinical feasibility and significance of using this approach.

The current study is innovative in two key aspects: (1) it is the first to validate the use of pose estimation for examining the kinematics of falls, and (2) it provides a foundation for future applications to analyze video-captured real-life falls. While OpenPose has been validated for gait^{17,37} and industrial behavior analysis^{38,39}, analyzing falls is inherently more complex. Falls involve rapid, dynamic, and unpredictable movements, often resulting in blurry video images, which create challenges for applying pose estimation. In contrast, gait and industrial tasks (e.g., squat task, lifting motion) involve slower, more predictable movements that are easier to track.

Additionally, falls among older adults are a significant public health concern. Our study demonstrates that pose estimation algorithms like OpenPose can be applied to extract meaningful data from video-captured falls, laying the groundwork for automatically detecting and classifying real-life falls captured by surveillance cameras. This approach will help overcome the limitations of traditional fall research, which relies on simulated, predictable falls in lab-based settings⁸. Furthermore, recently, emerging research has focused on using pose estimation algorithms to detect or predict real-life falls^{20,21}. While these studies show promising results in accurately detecting falls, they primarily rely on body position data extracted from pose estimation. Our study found that falling velocity obtained from pose-estimation video analysis is accurate enough to be useful. Adding this information could improve fall detection by not only detecting falls more accurately but also assessing their injury risk, enabling faster medical response if needed.

This study has several limitations. First, we captured video from an optimal viewpoint by setting the camera angle perpendicular to the participant, ensuring the participant was centered in the view. However, the accuracy of the pose estimation algorithm is influenced by various camera settings, such as camera angles, the participant's position in the field of view, other people in the view, and occlusion by other objects^{16,17,22,40}. Moreover, the frame rate and resolution of the videos used in our analysis (30 Hz, 1920 × 1080 pixels) were standard settings for household electronic cameras, smartphones, or tablets¹⁷. However, our results may not fully reflect the accuracy obtainable with videos of lower or higher frame rates or resolutions. Furthermore, to convert pixels to meters, we used an anthropometric reference based on the shoulder-to-toe length, estimated from the participant's height. However, this method might introduce errors in calculating kinematic parameters. Additionally, two major limitations of our statistical approach represent key constraints in this secondary analysis: (1) the inability to conduct repeated-measures analysis due to unequal trial numbers among the independent variables (e.g., falling strategy and falling side), and (2) the lack of sample size justification. While a balanced repeated-measures design would be preferable for comprehensively assessing the effects of individual fall strategies and fall sides, we conducted a two-way ANOVA (Fall Strategy × Fall Side) due to imbalances in the frequency of fall strategies and sides. This limitation arises from the secondary nature of the analysis, as the initial study was designed to observe participants' natural falling responses and involved a greater number of trials on the right side²⁵. Future studies should aim to standardize the number of trials across both fall sides and strategies to enable repeated-measures or mixed-effects analyses, allowing for a more robust evaluation of individual fall strategies and side effects. Secondly, the sample size for this study was not determined through a priori power calculation. However, a post hoc power calculation based on the mean absolute error (MAE) of hip impact acceleration yielded an effect size of $d = 0.69$ for differences across falling strategies. This calculation indicated that a sample size of 30 participants would be required to achieve 95% statistical power at a 5% significance level to detect differences in validation metrics across movement strategies during falls. Also, our fall experiments, which used the 'tether-release' method to simulate sideways falls on to a thick crash pad, may not accurately represent the biomechanical components of real-life falls, which typically involve both initial velocity and greater peak accelerations upon impacting rigid flooring²⁵. Another limitation of the study is that the study only investigated sideways falls, which limits generalizability of the results, given that fall direction can impact injury risk⁴¹.

Future studies should incorporate various fall directions (e.g., forward, backward falls), and compute diverse kinematic parameters (e.g., horizontal velocity, kinematics of other body parts) to enhance the generalizability of this method. Also, comparing performance of different AI-based pose estimation algorithms (e.g., DeepLabCut⁴², Alpha Pose⁴³, Uni Pose⁴⁴) applied to the same fall video data would be beneficial for advancing the field. Finally, exploring machine learning approaches to improve acceleration estimation, conducting a pilot study using real-world fall videos, and investigating the potential of transfer learning or fine-tuning the OpenPose model for fall detection could significantly enhance the study's contributions⁴⁵.

Conclusion

To our knowledge, this is the first study to validate the use of a pose estimation algorithm (OpenPose) for identifying kinematic parameters (i.e., hip impact velocity and acceleration) in video-captured falls among older adults. Furthermore, this study is among the few that have utilized older adults in falling experiments, who exhibited diverse movement patterns during a fall. The results showed that while OpenPose estimated hip impact velocity with good accuracy, the estimation of hip impact acceleration requires further improvement. Future validation studies involving diverse camera settings, fall contexts, and biomechanical parameters are warranted. This study has demonstrated the potential use of the pose estimation algorithm to accurately and efficiently extract time-series biomechanical parameters from real-life video-captured falls.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

RM contributed as the primary author with data analysis and manuscript drafting. YM and TB contributed to data analysis and manuscript drafting. YM, SNR, and JJS contributed to study design and manuscript editing. YM acted as lab PI.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Y.M.

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